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Palladium-Catalyzed Intermolecular [5 + 2] Cycloaddition: Highly Regioselective Synthesis of Allene-Containing 2-Benzoxepines

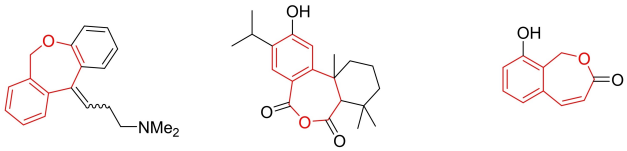
Bingyu Shen,[a, b] Aimei Yang,[b] Xihong Lu,[a] Zhijun Xin,*[a] and Xuehu Li*[a]

Palladium-catalyzed intermolecular [5 + 2] cycloaddition of propargylic carbonates with benzyl allenyl ethers has been developed for the synthesis of allene-containing 2-benzoxepines. The reaction proceeds by a pathway involving decarboxylative allenylpalladium formation, regioselective intermolecular insertion, and intramolecular nucleophilic attack. In this way, the sequential formation of C–C and C–O bonds results in the generation of a 2-benzoxepine skeleton in a one-pot process.

Introduction

As representative seven-membered oxygen-containing heterocycles, 2-benzoxepines represent a significant structural motif in numerous natural products and synthetic pharmaceuticals that exhibit a wide range of biological and physiological activities (Figure 1).[1] For example, Doxepin demonstrates both antidepressant and sedative-hypnotic effects,[2] Obtuanhydride displays antifungal activity,[3] whereas xylarinols A has proven to inhibit the activity of yeast α-glucosidase.[4] In view of the importance of 2-benzoxepines, the establishment of efficient and straightforward construction methods for synthesizing these scaffolds has attracted considerable attention from the organic synthesis community.[5]

In the past decades, tremendous advances have been achieved in synthetic methods for the construction of 2-benzoxepine skeletons, such as transannular macrocyclization via intramolecular *B*-alkyl Suzuki reaction,[6] tandem S_N2 alkylation/Michael addition[7] and transition-metal-catalyzed [2 + 2 + 2] cyclotrimerization,[8] coupling annulation,[9] [5 + 2] cycloaddition.[10] Nevertheless, the synthesis of seven-membered heterocycles continues to encounter significant challenges, primarily due to the unique thermodynamic and kinetic properties of seven-membered nitrogen/oxygen-containing heterocyclic skeletons, along with their own specific transannular interactions.[11] Consequently, the development of novel and efficient synthetic strategies to access 2-benzoxepine skeletons is highly desirable.



Doxepin Obtuanhydride Xylarinol A

Figure 1. Representative biologically or pharmacologically active molecules containing 2-benzoxepine skeletons.

Intermolecular cycloadditions of allenes derivatives via palladium-catalyzed have been discovered to efficiently integrate various structural fragments while maintaining high atom and step economy. Consequently, this method offers straightforward and effective approaches for the synthesis of heterocycles.[12] Remarkably, Ma *et al.* employed the aforementioned strategy to develop a series of novel methods for synthesizing complex six-membered oxygen-containing[13] and eight-membered nitrogen-containing[14] heterocycles, utilizing allene derivatives and propargylic carbonates as starting materials. However, the synthesis of seven-membered heterocycles has not been reported to date. Inspired by these findings, we envisioned a strategy for the construction of 2-benzoxepine skeletons through decarboxylation/cycloaddition of propargylic carbonates with benzyl allenyl ethers. Herein, we present our efforts on Pd(0)-catalyzed intermolecular [5 + 2] cycloaddition, which provides an ideal and concise approach for synthesizing functionalized 2-benzoxepine frameworks Scheme 1.

Results and Discussion

At the outset of our investigation, propargylic carbonate **1a** and benzyl allenyl ethers **2a** were chosen as the model substrates to determine the optimal experimental conditions. The results are summarized in Table 1. Using $Pd(OAc)_2$ as the catalyst, Tri(2-furyl)phosphine (TFP) as the ligand, Na_2CO_3 as the base and DMF as the solvent (at 100 °C under Ar for 0.5 h), gave 21% isolated yield of the desired compound (entry 1). To improve the reaction yield, various palladium catalysts, includ-

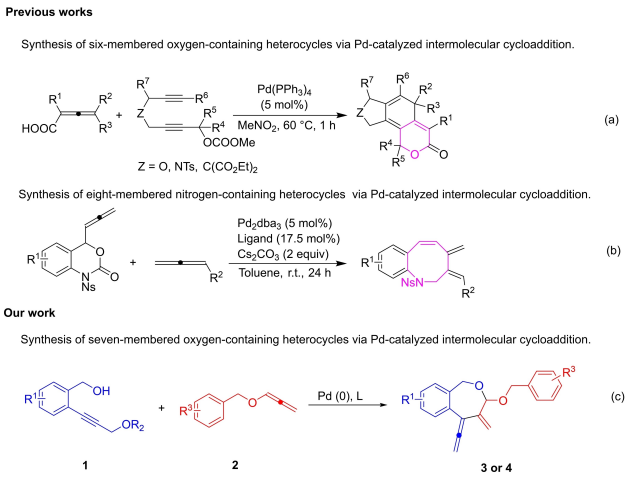
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Scheme 1. Several strategies for the synthesis of heterocycles via Pd-catalyzed intermolecular cycloaddition of propargylic carbonates and allene derivatives.

ing $Pd(OAc)_2$, $Pd(PPh_3)_4$, $Pd_2(dba)_3$, $Pd(dba)_2$ and $Pd(dba)_3{\cdot}CHCl_3$, were preferentially screened (entries 2–5). The results demonstrated that $Pd(PPh_3)_4$ is the most effective palladium source in this protocol, affording the target product **3a** in 39% yield (entry 2). Subsequently, we evaluated the different types of ligands necessary for the reaction (entries 6–13), such as monodentate phosphine ligands, bidentate phosphine ligands and boron reagents. The results revealed that monodentate phosphine ligands exhibited superior performance compared to both bidentate phosphine ligands and boron reagents. Notably, when 2-(diphenylphosphino) pyridine (L1) was employed as the ligand the yield of the target product increased to 66% (entry 9). In the absence of additional ligands, the yield of **3a** was limited to 32% (entry 14). Consequently, $Pd(PPh_3)_4$/L1 was confirmed as the optimal catalytic system for the cycloaddition reaction. Among the bases we investigated, Na_2CO_3 proved to be the most effective (entry 9), followed by K_2CO_3 (entry 15). In contrast, the target products were not detected when using Cs_2CO_3 or NEt_3 as the base (entries 16 and 17). Further screening of the temperature did not result in higher yields (entries 18 and 19), indicating that 100 °C is the optimal temperature (entry 9). Based on the aforementioned results, the optimal reaction conditions were established as follows: $Pd(PPh_3)_4$ (5 mol%), L1 (10 mol%), Na_2CO_3 (2.0 equiv), DMF (5 mL), at 100 °C under Ar for 0.5 h.

With the optimized reaction conditions in hand, we explored the scope and generality of the cyclization reaction through a systematic investigation of the electronic nature of the substituents on substrates **2** under the optimal conditions (Table 2). The yield of the *para*-methoxy-substituted product **3b** was 62%, which was slightly lower than that of product **3a**. Additionally, the yield of the *meta*-methoxy-substituted analogue **3c** further decreased to 55%. As the number of methoxy substituent groups on the benzene ring of substrates **2** increased, the yields of the corresponding multi-methoxy-substituted products **3d** and **3e** exhibited a downward trend. The substrate **2** containing dioxygen substituent demonstrated tolerance and afforded the desired cyclization product **3f** in 54% yield. Notably, the naphthylmethyl allenyl ether was successfully converted into **3g**, achieving a yield of 72%, which surpassed that of **3a**. The increase in yield may be attributed to the electron effects of the naphthalene ring. Next, we investigated the reactivity of benzyl allenyl ethers with fluorine or chlorine substituents, successfully obtaining the target products **3h** and **3i** in 43% and 55% yields, respectively. However, the iodine-substituted substrate **2j** did not give the target product, which could be primarily attributed to the oxidative insertion of Pd into the C–I bond, leading to an unintended coupling reaction.[15] To our surprise, substrates 2 containing a strong EWG (CN, CF_3) also furnished the expected products **3k** and **3l** in satisfactory yields. In summary, substrates **2** bearing EDG or EWG are well tolerated, delivering corresponding products in moderate to good yields.

To further evaluate the robustness of the methodology, the reactivity of dioxygen-substituted propargylic carbonate **1b** and diverse benzyl allenyl ethers **2** was investigated (Table 3). The catalytic system similarly demonstrated tolerance to the reaction of **1b** with benzyl allenyl ethers **2** which were substituted with methoxyl, fluorine, chlorine, trifluoromethyl and cyano groups, as well as with naphthylmethyl allenyl ether, yielding 2-benzoxepines in 35% to 60% (**4a–4h**). A comparison of the yields of **3a** and **4a**, **3b** and **4b**, **3d** and **4c**, **3g** and **4d**, **3h** and **4e**, **3i** and **4f**, **3k** and **4g**, as well as **3l** and **4h** revealed that, when reacting with the same benzyl allenyl ether **2**, the yields of 2-benzoxepines synthesized from substrate **1b** were lower than those synthesized from substrate **1a**. This observation indicated that the presence of electron-withdrawing substituents on the benzene ring of substrates **1** was not favorable for the reaction. As anticipated, benzyl allenyl ethers with a methyl substituent at the *ortho* or *para* position of the allenyl ether group underwent transformations smoothly, generating the corresponding products in 57% and 58% yields, respectively (**4i** and **4j**). However, due to steric hindrance, the yield of the isopropyl-substituted product **4k** was limited to only 44%.

Following the above experiments, we devoted our attention to various leaving groups on propargylic carbonates (Table 4). The reactions involving propargyl ethyl carbonate **1c** and propargyl benzyl carbonate **1d** demonstrated good compatibility and yielded acceptable results under the tested conditions (entries 1 and 2). In contrast, propargyl phosphate **1e** exhibited lower reactivity, potentially due to the reduced leaving ability of phosphate compared to that of the aforementioned carbonates (entry 3).

Based on previous studies[13] and experimental results, a plausible mechanism for the formation of functionalized 2-benzoxepine scaffolds is depicted in Scheme 2. Initially, oxidative addition of **1a** with catalytically active palladium(0) leads to afford allenylpalladium intermediate **5**.[16] Subsequently, species **5** undergoes highly regioselective intermolecular insertion of the nonterminal C=C bond on the allene moiety in **2a** to generate allylpalladium intermediate **6**.[17] In the presence of base, intermediate 6 is converted to intermediate **7**. Finally, the key intermediate **7** undergoes regioselective intramolecular

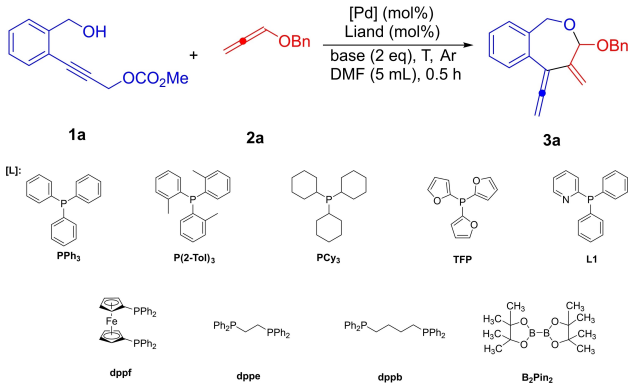
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Table 1. Optimization of the reaction conditions.[a]



Entry	Catalyst (mol%)	Ligand[b] (mol%)	Base (2 equiv.)	T (°C)	Yield (%)[c]
1	$Pd(OAc)_2$	TFP	Na_2CO_3	100	21
2	$Pd(PPh_3)_4$	TFP	Na_2CO_3	100	39
3	$Pd_2(dba)_3$	TFP	Na_2CO_3	100	26
4	$Pd(dba)_2$	TFP	Na_2CO_3	100	17
5	$Pd(dba)_3{\cdot}CHCl_3$	TFP	Na_2CO_3	100	N.D.
6	$Pd(PPh_3)_4$	PPh_3	Na_2CO_3	100	56
7	$Pd(PPh_3)_4$	$P(2-Tol)_3$	Na_2CO_3	100	30
8	$Pd(PPh_3)_4$	PCy_3	Na_2CO_3	100	41
9	**$Pd(PPh_3)_4$**	**L1**	**Na_2CO_3**	**100**	**66**
10	$Pd(PPh_3)_4$	Dppf	Na_2CO_3	100	N.D.
11	$Pd(PPh_3)_4$	Dppe	Na_2CO_3	100	5
12	$Pd(PPh_3)_4$	Dppb	Na_2CO_3	100	11
13	$Pd(PPh_3)_4$	B_2Pin_2	Na_2CO_3	100	34
14	$Pd(PPh_3)_4$		Na_2CO_3	100	32
15	$Pd(PPh_3)_4$	L1	K_2CO_3	100	44
16	$Pd(PPh_3)_4$	L1	Cs_2CO_3	100	N.D.
17	$Pd(PPh_3)_4$	L1	NEt_3	100	N.D.
18	$Pd(PPh_3)_4$	L1	Na_2CO_3	80	45
19	$Pd(PPh_3)_4$	L1	Na_2CO_3	120	60

[a] Conditions: **1a** (0.5 mmol, 1.0 equiv), **2a** (1.0 mmol, 2.0 equiv), $Pd(PPh_3)_4$ (5 mol%), L1 (10 mol%), Na_2CO_3 (2.0 equiv), DMF (5 mL), 100 °C, 0.5 h, under argon. [b] PPh_3: triphenylphosphine, $P(2\text{-Tol})_3$: tri(*o*-tolyl)phosphine, PCy_3: tricyclohexylphosphine, TFP: tri(2-furyl)phosphine, L1: 2-(diphenylphosphino)pyridine, dppf: 1,1′-bis(diphenylphosphino)ferrocene, dppe, 1,2-bis(diphenylphosphino)ethane, dppb: 1,4-bis(diphenylphosphino)butane, B_2Pin_2: bis(pinacolato)diboron. [c] Isolated yield. N.D. = not detected.

nucleophilic attack to form seven-membered ring product **3a** and regenerate the catalytically active palladium (0).[18] Additionally, electronic effects of the benzyloxy group in substrates 2 may be the key controlling factor that produces regioselectivity.

Conclusions

In summary, we have discovered a hitherto undescribed method for the synthesis of allene-containing 2-benzoxepines via palladium-catalyzed intermolecular [5 + 2] cycloaddition. Notably, this novel protocol features easy substrates availability, broad substrates scope, decent yields and excellent regioselectivity which demonstrates its great potential for the construction of complex and challenging seven-membered oxygen-containing heterocycles.

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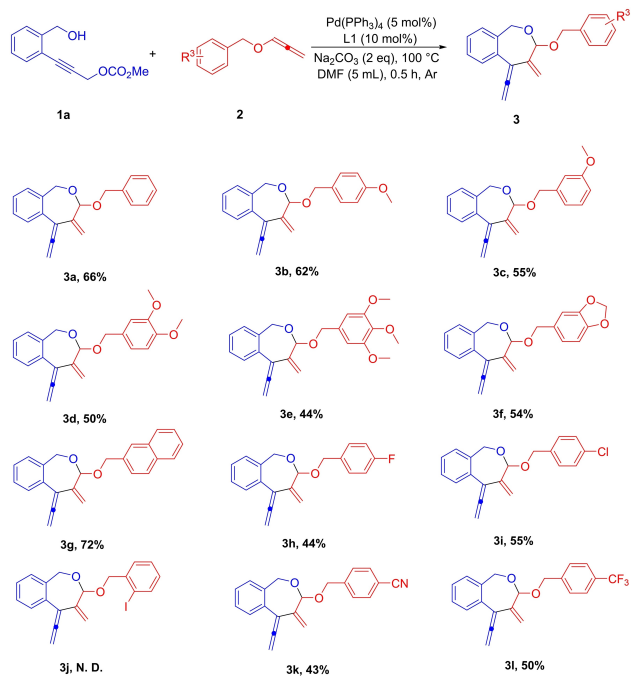
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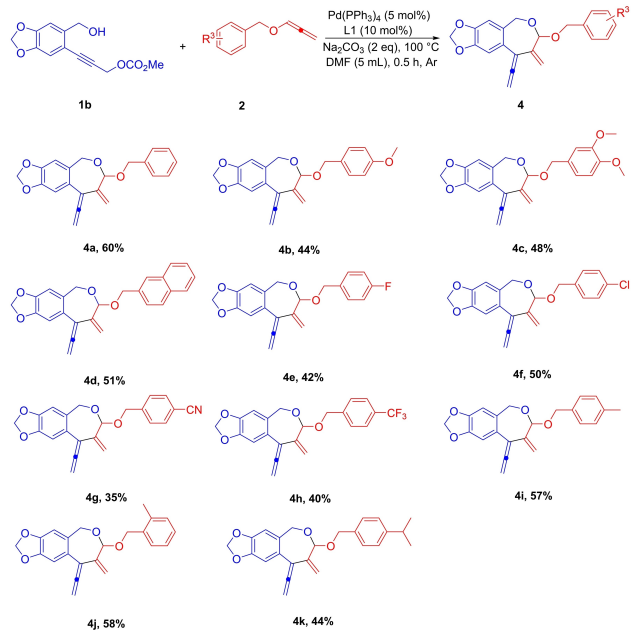
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Table 2. Scope of benzyl allenyl ethers **2** reacting with propargylic carbonate **1 a**.[a]



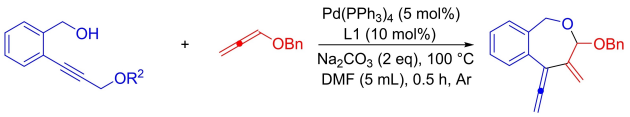
[a] Reaction conditions: **1 a** (0.5 mmol, 1.0 equiv), **2 a** (1.0 mmol, 2.0 equiv), $Pd(PPh_3)_4$ (5 mol%), L1 (10 mol%), Na_2CO_3 (2.0 equiv.), DMF (5 mL), 100 °C, 0.5 h, under argon. Isolated yield. N.D. = not detected.

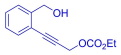
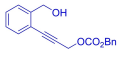
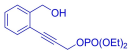
Table 3. Scope of benzyl allenyl ethers **2** reacting with propargylic carbonate **1 b**.[a]



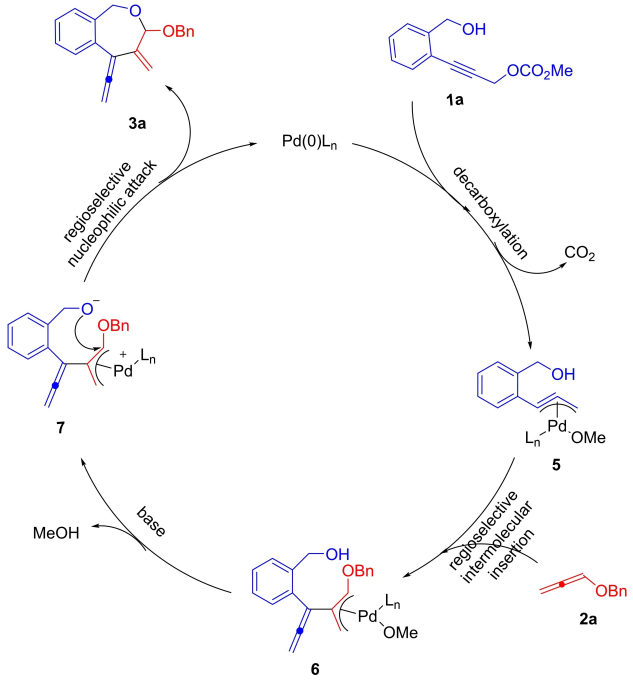
[a] Reaction conditions: **1 a** (0.5 mmol, 1.0 equiv), **2 a** (1.0 mmol, 2.0 equiv), $Pd(PPh_3)_4$ (5 mol%), L1 (10 mol%), Na_2CO_3 (2.0 equiv.), DMF (5 mL), 100 °C, 0.5 h, under argon. Isolated yield.

Table 4. Scope of propargylic esters **1** reacting with benzyl allenyl ethers **2 a**.[a]



Entry	1	2	Product	Yield[b] (%)
1	 **1 c**	**2 a**	**3 a**	51
2	 **1 d**	**2 a**	**3 a**	44
3	 **1 e**	**2 a**	**3 a**	38

[a] All reactions were carried out under the optimal conditions reported in the text. [b] Isolated yield.



Scheme 2. Proposed reaction mechanism.

Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

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